

A Measured Decoder-Only Transformer

Learning, inference, and compression trade-offs on a laptop-scale budget

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Abstract

We study whether a compact decoder-only Transformer can learn held-out scientific prose while exposing measurable systems trade-offs. The implementation includes byte-level tokenization, causal self-attention, RMSNorm, SwiGLU, tied embeddings, deterministic evaluation, autoregressive generation, KV caching, and a portable weight-only INT8 reference. The primary run uses approximately 1 MB of public-domain text from Darwin and Einstein, one fixed seed, and CPU measurements. It reaches 17.58 held-out perplexity on 112,111 next-byte targets. KV caching reduces the recorded 96-token decode median from 119.4 ms to 33.4 ms. INT8 reduces stored model tensors by 53.6% but is slower in this reference implementation.

These results establish a reproducible laptop-scale baseline. They do not establish broad language ability, scientific understanding, statistical significance, or Apple M4/MPS performance.

Headline measurements

Measure	Result	Protocol
Held-out perplexity	17.58	112,111 next-byte targets
KV-cache speedup	3.58x	CPU; seven timed repeats
Stored tensor reduction	53.6%	MLP weight-only INT8
CI verification	16 tests pass	Ruff + pytest on macOS CI

Code, raw JSON, corpus hashes, figures, and reproduction commands: github.com/AnvitDevadiga/MiniLM-Lab

1 Introduction

Small language-model repositories often demonstrate a forward pass or a training script, but leave the measurement protocol implicit. MiniLM Lab treats the model as a systems object: the corpus, split, optimizer steps, evaluator, cache path, and compression path are explicit. The goal is not to compete with large language models. The goal is to make the full byte-to-generation path inspectable and falsifiable on ordinary hardware.

2 Research questions

(Q1) Does the compact model reduce held-out next-byte loss during training on a reproducible science corpus?

(Q2) Does reusing past key/value states reduce autoregressive decode latency under a fixed CPU protocol?

(Q3) What is the quality, storage, and latency trade-off of a portable weight-only INT8 reference?

3 Data and protocol

The corpus consists of Project Gutenberg editions of Darwin's *On the Origin of Species* and Einstein's *Relativity: The Special and General Theory*. Boilerplate is removed. Approximately the first 90% of each document is used for training; the remainder is held out at a paragraph boundary. Source URLs, raw SHA-256 hashes, prepared split hashes, and byte counts are stored in `data/science/manifest.json`.

Protocol item	Published value	Interpretation
Training bytes	1,008,977	Approximately 1 MB
Held-out bytes	112,112	Same books; later passages
Seed	42	One reported run
Optimizer updates	1,000	Four evaluation checkpoints
Tokenizer	UTF-8 bytes	256-value vocabulary

4 Model architecture

The model is a decoder-only Transformer with four blocks, embedding width 192, six attention heads, context length 128, learned positions, RMSNorm, SwiGLU feed-forward layers, residual connections, and tied input/output embeddings. Training uses AdamW with learning rate 3e-4, microbatch size 8, four-way gradient accumulation, and seed 42.

5 Evaluation

The evaluator scores every next token once in non-overlapping context windows. The first token of each window supplies left context. Perplexity is $\exp(\text{mean negative log-likelihood})$. Bits per byte is computed from the byte-normalized loss and the exact scored-byte denominator.

6 Implementation controls

Checkpoint selection uses deterministic full-split validation. Resume checkpoints include model state, optimizer state, CPU RNG state, configuration, corpus hash, and metrics. Cache outputs are checked against full-forward outputs for learned positions and RoPE.

7 Parameterization

Component	Configuration
Attention	Causal MHA
Norm	RMSNorm
Activation	SwiGLU
Position	Learned / RoPE
Output	Tied embeddings

8 Architecture flow

Stage	Representation	Purpose
Input	UTF-8 bytes	Corpus interface
Transformer	Causal hidden states	Next-byte prediction
Output	Vocabulary logits	Autoregressive sampling

9 Results: learning behavior

The held-out curve decreases at the final checkpoint. Because only four checkpoints and one seed are reported, the figure is descriptive rather than a statistical comparison.

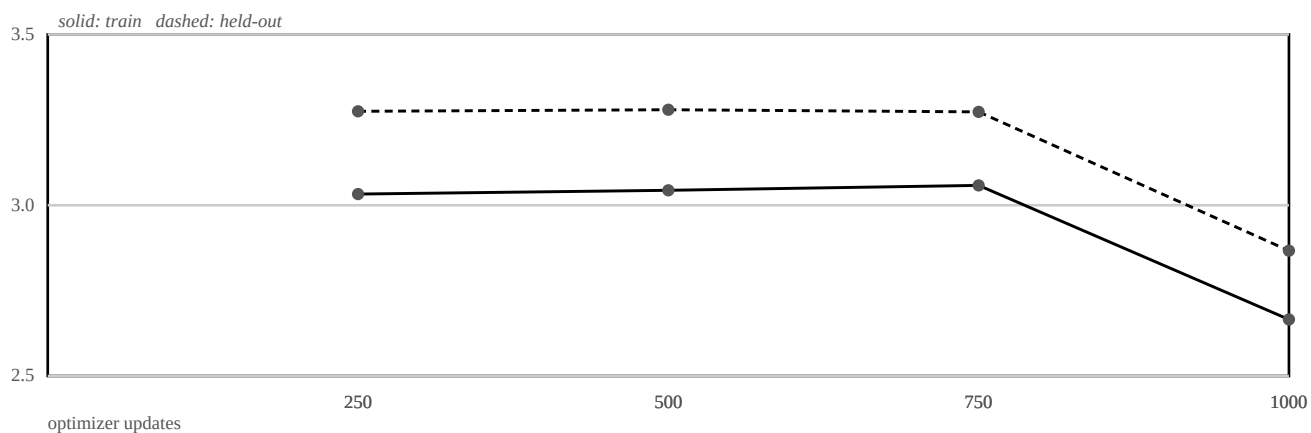


Figure 1. Training and held-out loss across four optimizer checkpoints. The held-out series covers 112,111 targets.

Metric	Value	Definition
Validation loss	2.8669 nats/token	Mean held-out negative log-likelihood
Perplexity	17.58	$\exp(\text{validation loss})$
Bits per byte	4.136	Byte-normalized loss
Targets	112,111	Unique next-byte targets

10 Per-source evaluation

Source	Targets	Perplexity	Bits/byte
Darwin	93,405	17.76	4.150
Einstein	18,703	16.68	4.060

These are within-book continuation tests, not unseen-author generalization tests.

11 Results: inference and compression

For the cache benchmark, prompt, sampling settings, generated length, warm-up count, and repeat count are fixed. Bars show the mean of seven timed runs; whiskers show plus or minus one standard deviation.

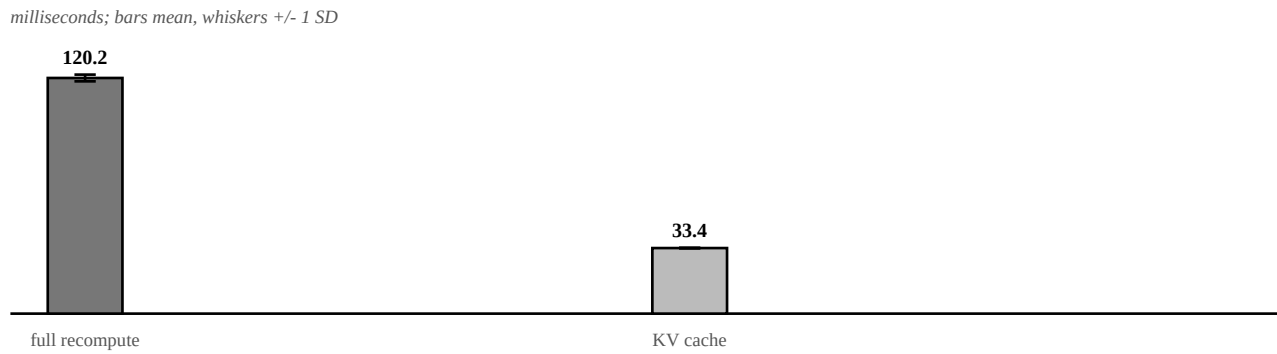


Figure 2. Decode latency for 96 generated tokens after a 21-token prompt. The cached path is 3.58x faster by median on the recorded CPU run.

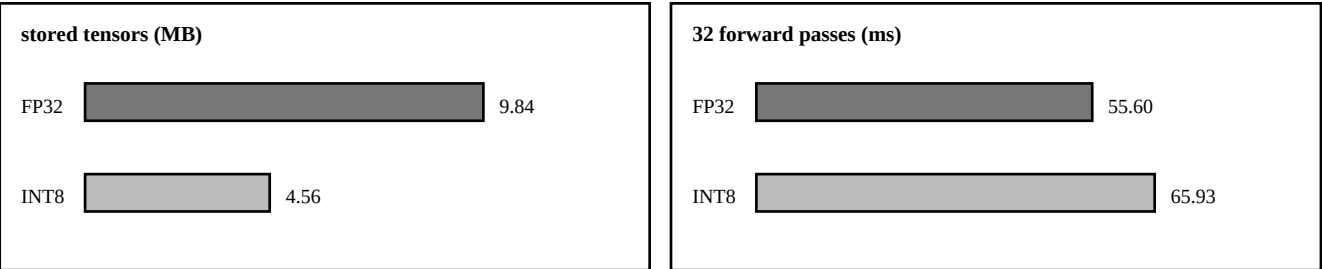


Figure 3. Portable weight-only INT8 reference. Storage decreases while the unoptimized dequantization path increases latency.

Measure	FP32	INT8	Difference
Stored tensors	9.84 MB	4.56 MB	-53.6%
Validation loss	2.866917	2.867067	+0.000150
32 forward passes	55.6 ms	65.9 ms	INT8 slower

12 Discussion and limitations

The evidence supports a compact, inspectable language-model systems baseline. It demonstrates the full path from public text to model training, held-out scoring, autoregressive generation, cache reuse, and a compression trade-off. It does not support claims of broad language ability, scientific understanding, production serving, or superiority over larger open-source training systems.

13 Threats to validity

Threat	Consequence
One seed	No uncertainty estimate for model quality or method superiority.
One model shape	No scaling-law or architecture comparison.
Within-book holdout	Style and repeated phrases may cross the split.
Small corpus	Perplexity does not imply general knowledge.
CPU environment	Does not establish Apple M4/MPS performance.
Reference INT8 path	Storage result is not accelerated deployment evidence.

14 Required next experiments

A stronger flagship study should add three or more matched seeds, a third-author test book excluded from training, an M4 MPS run recording throughput and memory, matched learned-position versus RoPE and context-length comparisons, and an accelerated quantization backend where hardware supports it.

15 Conclusion

MiniLM Lab is valuable as an evidence-first engineering study. Its differentiator is the explicit connection between implementation, protocol, raw measurements, tests, and limitations. The M4 benchmark and multi-seed/unseen-author experiments are the remaining work required for a stronger research claim.

16 Reproducibility and references

The repository includes the implementation, 16 correctness tests, pinned corpus manifest, raw experiment record, protocol, generated figures, and this report. The published run can be reproduced with the commands in README.md.

17 Reference paper used as a pattern

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. arXiv:1810.04805v2. The reference informed the separation of objective, architecture, protocol, results, and limitations. MiniLM Lab does not claim BERT-like scale, capability, or benchmark performance.

18 Data sources

Project Gutenberg eBook #1228, Darwin's On the Origin of Species, and eBook #30155, Einstein's Relativity: The Special and General Theory. URLs and SHA-256 hashes are recorded in data/science/manifest.json.